

ThoxMini Air User Manual

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THOX.ai

Welcome

Thank you for backing ThoxMini Air. ThoxMini Air takes the compute core of ThoxMini, drops in a small LiPo cell for tetherless operation, and adds a magnetic ring on top so each unit clicks neatly onto the next. That last detail is the headline: a stack of Airs becomes a self-cooled compute cluster you can carry in one hand.

What is in the box

- 1 x ThoxMini Air node in v4 enclosure (matte black back, light gray halo ring)
- 1 x USB-A to USB-C cable, 1 m
- 1 x quick-start card
- 1 x THOX brand sticker
- 1 x spec card with the device serial number
- 1 x Liter 602530 500 mAh LiPo cell (pre-installed and pre-charged)

Safety highlights

- The LiPo cell is pre-installed and connected. Do not pry the enclosure open. The cell is small (500 mAh) but a punctured LiPo can vent flame.
- Do not charge in an enclosed bag overnight. ThoxMini Air has overcharge protection but treat any LiPo with respect.
- Operating range is 0 C to 45 C ambient.
- The magnetic halo ring is strong (N52). Keep it 100 mm from credit cards, hard drives, mechanical watches, and medical implants.
- The enclosure is not weather-sealed. Keep it dry.

What ThoxMini Air is for

- Tetherless private AI agent that runs for roughly 90 minutes off the internal cell on a typical workload
- All 14 ThoxMini skills plus the cluster-aware skill stub
- Cluster member: stack 4 to 8 Airs onto a MagStack Cluster Dock and the units self-discover and load-balance
- Same compute, same gateway, same web UI as ThoxMini

Quick start

1. **Unbox.** The cell is pre-installed and pre-charged.
2. **Single-tap the side button** to power on. The halo ring breathes white during boot. About 25 seconds.
3. **Plug the USB-C cable** into your computer (data + charge) or into any 5 V / 1 A USB power source for charge only.
4. **Open a browser** to `http://172.32.0.70:18790` (USB-connected) or `http://thoxmini-air.local:18790` (mDNS on your LAN).
5. **Send your first prompt.** Same agent and skills as ThoxMini.

To stack with other Airs: hold a second Air above this one until the two halo rings click. Repeat. The bottom-most Air handles power if sitting on a Cluster Dock; otherwise each Air uses its own cell.

Full setup

1. Compute side: follow the ThoxMini runbook

The Air variant shares its compute, OS, and agent layer with ThoxMini. For per-unit OS detail, identity, SSH key install, Tailscale, and the skill catalog, follow the **ThoxMini User Manual** section "Full setup" in full. Everything from "Confirm the device identity" through "Health check" applies one for one.

2. Confirm the cell is wired

```
ssh root@172.32.0.70
cat /sys/class/power_supply/battery/uevent
```

You should see `POWER_SUPPLY_STATUS`, `POWER_SUPPLY_CAPACITY`, and `POWER_SUPPLY_VOLTAGE_NOW`. If the file is missing, the JST-PH-2.0 cell pigtail came loose in shipping; contact `dev@thox.ai`.

3. Enable cluster discovery (only if you have more than one Air)

```
echo 'THOXMINI_CLUSTER=on' | tee -a /etc/thoxmini/thoxmini.env
systemctl restart thoxymicro
```

Each Air will broadcast on the USB-Ethernet link and on the tailnet (if joined). Within 10 seconds of cluster boot, the gateway on each Air exposes a `/cluster/peers` endpoint listing every reachable Air.

4. Stack onto a MagStack Cluster Dock

If you also have a MagStack Cluster Dock:

1. Place the Dock on a flat surface; the front spec plate faces you.
2. Set the first Air face-down so its halo ring drops into the well. Plug the USB-C cable from the manifold into the Air.
3. Stack the next Air on top; the halo rings snap together. Plug its USB-C cable. Repeat for 4 to 8 Airs.
4. The cluster powers from the USB-C hub the manifold cables run to.
5. Drop the cap over the topmost halo ring.

5. Halo ring behavior

Pattern	Meaning
Slow breathe white	Booting
Solid soft white	Idle, healthy
Solid cyan	Active inference

Slow pulse magenta	Talking to a peer in a cluster
Fast blink red	Fault; check <code>journalctl -u thoxymicro</code>
Off	Powered down or cell empty

6. Charging

Plug any 5 V / 1 A USB-C source. The halo ring stays soft white while charging and the gateway is still reachable. Full charge from empty takes about 90 minutes.

7. Hard power down

Hold the side button for 6 seconds. The halo ring fades to off. Use this before storing for more than a week.



Troubleshooting

Halo ring stays off after pressing the button

1. The cell may be flat. Plug a USB-C cable; the ring should breathe white within 5 seconds.
2. Hold the side button for 10 seconds to force a deep reset.
3. If still nothing, contact `dev@thox.ai`. Do not pry the case open.

Halo ring blinks red

```
ssh root@172.32.0.70
journalctl -u thoxymicro -n 200
```

Common causes: NPU not enumerated (reboot), OOM (switch model), disk full (`df -h /`).

Stack does not self-discover

1. Confirm `THOXMINI_CLUSTER=on` is set on every Air.
2. On each Air check `curl http://127.0.0.1:18790/cluster/peers`.
3. If peers are missing, the USB-Ethernet link between Airs may be broken; confirm the Cluster Dock USB-C manifold cables are seated in each Air.

Charging stalls partway

The cell stops at ~80% if the case is hotter than 40 C; this is the charge-management IC protecting the cell. Move to a cooler surface, or detach the bottom Air from a hot stack, and the charge resumes.

Two Airs will not snap together

The halo magnets are polarized: each ring has a "north" and "south" face. The factory orientation snaps every Air to the next. If you disassembled a stack and resealed a single ring backwards, flip it.

Factory restore

Same as ThoxMini:

```
ssh root@172.32.0.70
/opt/thox/bin/thoxmini-firstboot --reset
reboot
```

Warranty and support

Limited 1-year warranty

THOX.ai warrants ThoxMini Air against defects in materials and workmanship for one (1) year from the date of shipment. The warranty covers the enclosure, the magnet ring, the JST cell pigtail, and the compute board. It does **not** cover the LiPo cell beyond 6 months (consumable), drops, immersion, overvoltage, unauthorized modification, or use outside the documented operating conditions.

Make a claim by emailing `dev@thox.ai` with your order number, a photo of the issue, and the serial number from the spec card or `/opt/thox/etc/device.json`.

Support

- Email: `dev@thox.ai`
- Docs: <https://docs.thox.ai/thoxmini-air>
- Community: <https://github.com/ttracx/magstack-air/discussions>

Telemetry opt-out

Telemetry ships **off**. See ThoxMini manual for the same verification steps.

License summary

Identical to ThoxMini: GPL-2.0 (kernel + Ubuntu userland), BSD-3-Clause (Tailscale), Apache-2.0 (thoxymicro + agents + THOX models), MIT (llama.cpp + UI assets), Rockchip OEM (RKNN runtime). Full text on device at `/usr/share/doc/thoxmini-air/LICENSES.md`.

Open source notice

Component	License	Source
Ubuntu 22.04 (armv7l)	Various	ubuntu.com
Linux kernel 5.10 + Rockchip patches	GPL-2.0	rockchip-linux
systemd	LGPL-2.1	systemd.io
OpenSSH	BSD-2-Clause	openssh.com
adbd	Apache-2.0	android.googlesource.com
Tailscale	BSD-3-Clause	tailscale.com
thoxymicro (THOX)	Apache-2.0	github.com/ttracx/thoxymicro
magstack-air cluster crate (THOX)	Apache-2.0	github.com/ttracx/magstack-air
llama.cpp	MIT	github.com/ggerganov/llama.cpp
ThoxMicro-125M	Apache-2.0	github.com/ttracx/thox-micro-125m
ThoxLLM-327M-v2	Apache-2.0	huggingface.co/Thox-ai (private)
ThoxGem-E4B	Apache-2.0 (Gemma terms)	thox-gemma4 Phase B
RKNN runtime	Rockchip OEM	rockchip-linux/rknpu

The v4 enclosure design is published under Apache-2.0 at [ttracx/thox-3dprint-kit/devices/thoxmini-air](https://github.com/ttracx/thox-3dprint-kit/tree/main/devices/thoxmini-air). The MagStack ring geometry is filed under IP-013; the printed reference design is permissively licensed.

This product complies with FCC Part 15 Class B (declaration on file) and CE marking (declaration on file). The LiPo cell ships in the 50%-charged state per IATA PI 967 Section II air-shipment rules. Recycle exhausted cells at any cell-recycling drop-off (Call2Recycle in the US; municipal hazmat in the EU). Do not throw in household trash.

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